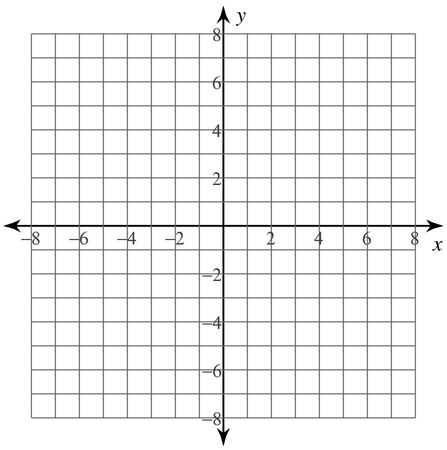
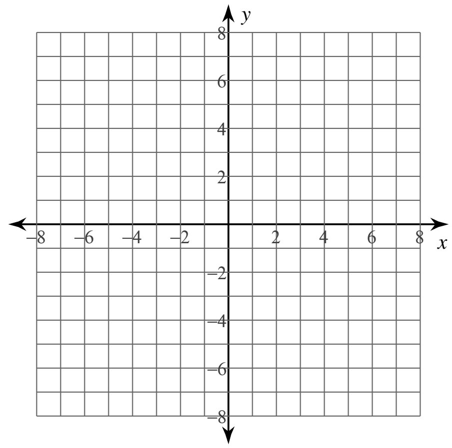
For each problem, find the area of the region enclosed by the curves. You may use the provided graph to sketch the curves and shade the enclosed region.

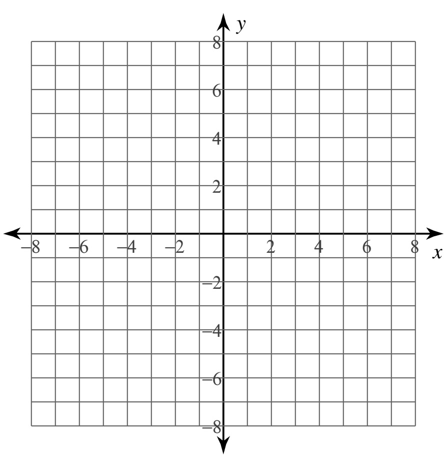
5) $y = -2x^2 - 1$
$y = -x + 3$
$x = 0$
$x = 1$



6) $y = 2\sqrt[3]{x^2}$
$y = x$



7) $y = -x^3 + 6x$
$y = -x^2$



8) $y = -2 \cdot \sec^2 x$
$y = 2\cos x$
$x = 0$
$x = \frac{\pi}{4}$